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DRCFusion: dual-stream radar-camera fusion for object detection in adverse weather conditions

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ABSTRACT
Robust object detection is crucial for the safety of autonomous driving. Cameras and radar sensors work in synergy as they provide complementary sources of information. Existing methods mostly use independent dual-branch frameworks to generate Bird's Eye View (BEV) images from radar and camera data, and then perform adaptive modality fusion. However, in adverse weather conditions, merging the features of both sensors presents challenges because cameras are significantly affected by weather. How to dynamically adjust the weight of camera features and guide the generation of camera BEV images has a substantial impact on the final detection results. In this paper, we propose DRCFusion, a radar-camera object detector. Specifically, we design the Semantic Transfer module, which adaptively adjusts the weight of camera image features based on varying weather conditions. To enhance the guidance for the camera in generating BEV features, a Geometric Transfer module is developed, which implicitly corrects the feature distribution of the camera image in the BEV space, utilising radar spatial features as guidance. We evaluate the performance of this method in object detection on the Radiate dataset. Comparisons with baseline methods demonstrate the performance improvements brought by the proposed approach.

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KEYWORDS
Object detection; adverse weather; multi-sensor fusion

1. Introduction

Autonomous driving systems rely on sensing technologies to ensure robust perception of dynamic targets, which is essential for reliable and safe vehicle detection (Hu et al., 2024; X. Lin et al., 2024; Liu & Ke, 2023). In recent years, target perception using camera-based systems has emerged as a prominent area of research. Cameras are advantageous due to their cost-effectiveness and their ability to capture intricate details such as color and texture, offering high-resolution semantic information critical for object detection. However, relying solely on a single camera sensor is insufficient for achieving high precision and robust object detection, as 2D images lack accurate depth information (Ma et al., 2021). Additionally, the performance of cameras can be severely affected under adverse weather conditions or low lighting, potentially leading to failures in perception tasks (Z. Liu et al., 2023).

To address these issues, many multimodal perception methods have focussed on the fusion between camera and radar sensors. Radar primarily operates at a frequency of 77 GHz, using millimeter waves to measure the distance, speed, and orientation of objects (Charvat, 2014). Due to these capabilities, radar can penetrate or reflect off small particles such as rain, fog, snow, and dust, enabling remote sensing under adverse weather conditions. Consequently, the fusion of cameras and radar can compensate for the limitations of individual sensors, providing a more comprehensive and reliable means of object detection (Bijelic et al., 2020).

Existing researches attempts to enrich radar features using camera visual data (Kim, Kim et al., 2023; Long et al., 2023; Zhou et al., 2023). Additionally, to improve camera depth estimation, current methods explicitly supervise the depth estimation process using radar information (Huang et al., 2021; Hwang et al., 2022; Y. Li et al., 2023). These methods have demonstrated favourable outcomes even under adverse weather conditions; however, as weather worsens further, images captured by cameras may lose clarity due to environmental effects. Using these blurred images indiscriminately to enhance radar features may degrade detection accuracy, as shown in Figure 1.

Additionally, vision-based BEV methods rely on depth supervision, which can be affected when camera images are unclear, significantly reducing the quality of depth estimation (Huang et al., 2021; Y. Li et al., 2023). To address this, we propose a radar-camera object detector called DRCFusion, which includes two key designs: the Semantic Transfer module for effectively extracting features from camera images and the

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Figure 1. The first row, from left to right, shows images captured by the camera and information collected by the radar on a clear day, demonstrating the correspondence between targets detected by different sensors. The second row, from left to right, displays images captured by the camera and radar on a snowy day. When the camera images are affected, the association between targets in the images and radar data is lost.

Geometric Transfer module for implicitly improving the projection of camera images onto the BEV plane in adverse weather conditions.

Specifically, the Semantic Transfer module introduces learnable control variable γ, which adaptively adjust the weight of the camera image features. Finally, we concatenate the enhanced radar features with the original radar features to obtain the fused radar features. In the Geometric Transfer module, we use a deformable attention module (Vaswani, 2017; Zhu et al., 2020), guided by radar spatial priors, to transform the camera images into BEV features, and perform feature aggregation adaptively using channel-wise and spatial fusion layers.

The main contributions of this work are summarised as follows:

- A novel DRCFusion object detection framework is proposed, which adopts a dual-stream architecture that dynamically interacts between image and radar features across the two branches, achieving robust object detection in adverse weather conditions.
- The Semantic Transfer module is proposed, which adaptively fuses image features during the integration process. It can adjust the weight of camera image features according to different weather conditions.
- To enhance the projection of camera images onto the BEV plane under adverse weather, the Geometric Transfer module is proposed, which implicitly corrects the feature distribution of camera images in BEV space under the guidance of radar spatial features.
- The model is evaluated on the RADIATE dataset, and a comprehensive comparison with baseline methods demonstrates the performance improvements brought by this model.

2. Related work

2.1. Camera-based 3D object detection

Cameras can provide rich texture features and visual information, but these naturally lack depth information, posing significant challenges for 3D object detection using cameras. In recent years, considerable effort has been made to infer pixel depth from images captured by cameras for 3D object detection. Additionally, transformer-based methods use attention mechanisms to map images to BEV representations and perform object detection based on the BEV (Vaswani, 2017). Overall, the methods can be classified into two main categories: depth-estimation-based methods and transformer-based methods.

Depth-estimation-based methods primarily utilises the method from the LSS (Philion & Fidler, 2020), which explicitly predicts the depth distribution of each pixel feature using a depth estimation network. It performs an outer product operation with the image features and projects the 3D features onto the BEV plane, incorporating the camera's intrinsic and extrinsic parameters. BEVDet builds on the viewpoint changes of LSS to further conduct 3D object detection in BEV features (Huang et al., 2021). BEVDepth introduces lidar information for depth supervision to optimise the network's depth estimation (Y. Li et al., 2023). Based on BEVDet, BEVDet4D spatially aligns BEV features from historical image frames, significantly improving the performance of speed prediction (Huang & Huang, 2022).

Transformer-based methods primarily construct queries and utilise attention mechanisms to identify corresponding image features, thereby mapping perspective views to either voxels or bird's eye view. BEVFormer integrates the spatio-temporal information from multi-camera images into a unified BEV format through the application of deformable attention mechanisms (Z. Li et al., 2022). SparseBEV combines a scale-adaptive self-attention module with an adaptive spatio-temporal sampling module to dynamically perceive BEV and temporal information (H. Liu et al., 2023). CLIP-BEVFormer leverages the method of contrastive learning to integrate the flow of ground truth annotations into the processes of BEV encoding and perceptual decoding, thereby enhancing the accuracy of environmental interpretation within autonomous driving systems (Pan et al., 2024).

2.2. Radar-based 3D object detection

Frequency modulated continuous wave (FMCW) radar typically offer two data representation forms: radio frequency (RF) images and radar points. RF images are generated from the raw radar signals through a series of fast Fourier transformations, encoding the sensor's readings. Radar points, on the other hand, are obtained from RF images using peak detection algorithms such as Constant False Alarm Rate (CFAR) (Richards et al., 2010). A drawback of radar points is that they have incomplete recall and lose the contextual information of the radar echoes, retaining only distance, azimuth, and Doppler information. Consequently, radar points are not suitable for effective monomodal object detection (Qi et al., 2017), which is why most works utilise this data format primarily to facilitate fusion (Y. J. Li et al., 2022; Z. Lin et al., 2024; Shamsfakhr et al., 2024; Wang et al., 2023). In contrast, RF images maintain rich environmental context, even complete object motion information, allowing deep learning models to comprehend the semantics of the scene (P. Li et al., 2022). Method (P. Li et al., 2022) harnesses the continuity of an object's existence and its attributes, including size and direction. It innovates by incorporating a temporal relation layer designed to exploit the sequential temporal dynamics across radar imagery frames, thereby facilitating the identification of targets. Based on the above discussion, we propose a radar RF image-based object detector that can produce reasonable object detection predictions from radar inputs.

2.3. Radar-camera 3D object detection

Radar is a popular sensor for 3D object detection in autonomous driving vehicles, characterised by its affordability and resilience to adverse weather conditions. Additionally, it offers long-range perception and Doppler velocity measurements. For millimeter-wave radars, the sparsity of data and the lack of rich semantic information typically relegate them to an auxiliary role in 3D object detection, where they are combined with images captured by cameras. Consequently, considerable research has been dedicated to exploring how to utilise image information to enrich radar features and effectively integrate the spatial features between the two modalities. Based on the different fusion methods, it can be classified into feature-level fusion and spatial-level fusion.

Feature-level fusion enriches radar features by leveraging image information. CenterFusion leverages a centre point detection network to yield preliminary 3D detection outcomes from imagery and utilises a conical framework for associating radar-based detections with the central points of corresponding objects, thereby facilitating subsequent refinement (Nabati & Qi, 2021). RCFusion introduces a radar PillarNet to fabricate radar pseudo-images and integrates a weighted fusion module designed to amalgamate the BEV features derived from both radar and camera sources (Zheng et al., 2023). RCBEV fuses the multi-modal features with a two-stage fusion model called point-fusion and ROIfusion that realizesthe feature-level fusion under the BEV (Zhou et al., 2023).

Spatial-level fusion aims to effectively align the spatial features of both modalities. CRN utilises a multimodal deformable attention mechanism to spatially align the BEV of images and radar, alleviating the problem of spatial misalignment between the two modalities (Kim, Shin et al., 2023). RCBEVDet presents an adeptly architected RadarBEVNet, dedicated to the efficient extraction of radar BEV features, and incorporates a cross-attention multi-layer fusion module to achieve robust multimodal feature alignment and integration (Z. Lin et al., 2024). In the context of radio frequency (RF) imagery, CramNet brings forth a ray-constrained cross-attention mechanism that serves to align features between camera and radar data (Hwang et al., 2022).

Although significant progress has been made in the field of multi-sensor fusion, existing methods still face limitations under adverse weather conditions such as rain, fog, and snow. Depth-estimation-based methods tend to suffer from inaccurate depth estimates when camera images are blurred, leading to a degradation in BEV projection quality. While Transformer-based methods can handle multi-modal data, they still heavily rely on camera features under adverse weather, making them vulnerable to environmental influences.

Compared to existing methods, the core advantage of the DRCFusion framework is its unique modular design. The Semantic Transfer module dynamically optimises the feature fusion process under different weather conditions by adaptively adjusting the weight

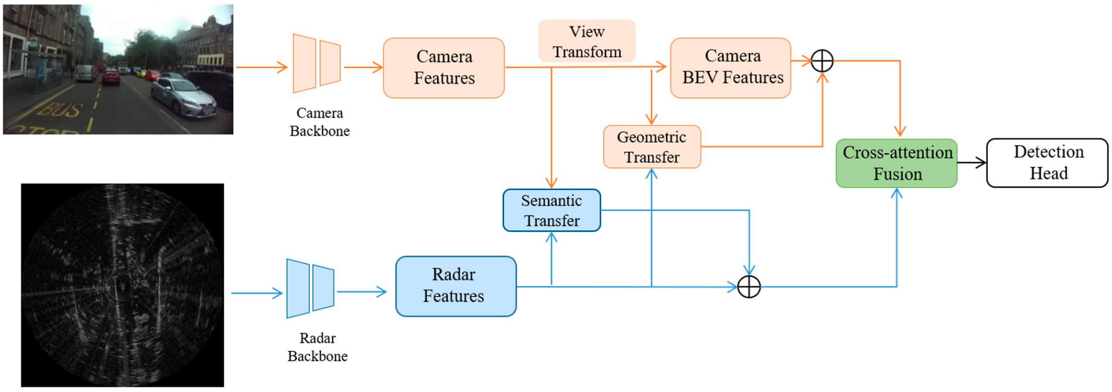

Figure 2. Overall pipeline of DRCFusion. Initially, features are extracted from radar and image data through their respective backbones. Subsequently, the Semantic Transfer module aggregates features from the Camera Features to enrich the Radar Features. Within the camera image sub-stream, images are projected onto the BEV plane, and the Geometric Transfer module is utilised to enhance the quality of the image BEV features. Finally, the cross-attention Fusion integrates the BEV features from both image and radar, and the fused features are employed for the object detection task.

of the camera image features. Additionally, the Geometric Transfer module utilises the spatial features of radar as guidance, implicitly correcting the feature distribution of the camera images in the BEV space, thereby significantly improving the accuracy of camera image projections under adverse weather conditions.

3. Methodology

Our proposed DRCFusion network is shown in Figure 2, where radar and image features are extracted separately through the Radar Backbone and Camera Backbone. Then, the Semantic Transfer module dynamically aggregates features from Camera Features to enrich Radar Features and adaptively adjusts the weight of the camera image features. In the camera image stream, the image features are first projected onto the BEV plane via View Transformation (Reading et al., 2021), while the Geometric Transfer module extracts accurate spatial information from the original radar features to improve the quality of the generated image BEV features. Finally, cross-attention Fusion aligns and integrates the BEV features from both image and radar, and the fused features are used for object detection tasks.

3.1. Semantic transfer

Under adverse weather conditions, the quality of camera images often deteriorates significantly, resulting in unreliable image features. Directly utilising these features may introduce noise, which can degrade detection accuracy. To address this challenge, we introduce the Semantic Transfer module, which adaptively adjusts the feature weights, dynamically balancing the contributions of camera and radar features. This approach ensures robust detection performance across adverse weather conditions.

Radar data has a single format, and features extracted solely from radar data are limited. However, images contain rich texture information, and leveraging image data to enrich radar features can improve object detection accuracy. For radar features with dimension d and size $h \times w$ we adopt an approach inspired by transfusion (Bai et al., 2022). Initially, we predict a heatmap $S \in \mathbb{R}^{w \times \times h \times n}$ for a specific category, where n denotes the number of classes, and the size of the heatmap is also $h \times w$. The top k local maxima within this heatmap are designated as the centroids of potential objects r_i. Employing a calibration matrix M, we project the positions of these potential objects r_i from the radar onto the image, establishing reference points p_i that facilitate the integration of data across modalities:

$$p_i = M \cdot r_i \tag{1}$$

where M is the product of the camera intrinsic matrix and the extrinsic matrix, and operation $\cdot$ is matrix multiplication.

Building upon the reference points, we generate aggregated image features F_I^i by applying learned offsets to the image features F_I and weighting a set of surrounding image features F_I^k in the vicinity of these reference points. We denote the features of potential objects on the radar as query Q_i, and the sampled features from the image as key K_i and value V_i. The entire dynamic fusion process can be represented as follows:

$$\begin{aligned} &DeformAtten(Q_i, p_i, F_I) \\ &= \sum_{m=1}^{M} W_m \left[\sum_{k=1}^{K} [A_{mik} \cdot W'_m F_I(p_i + \Delta p_{mik})] \right] \end{aligned} \tag{2}$$

In our formulation, W_m and W'_m are learnable weights, M is the number of self-attention heads and K is the total number of sampled points, which determines the scope of feature aggregation. Δp_{mik} and A_{mik} represent

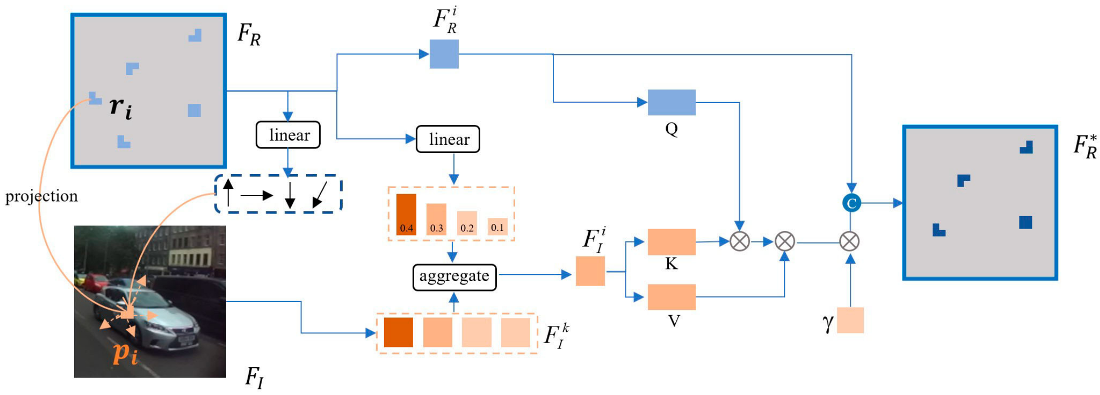

Figure 3. Semantic Transfer module. Initially, we forecast reference points on the radar. By leveraging a projection matrix, these points are projected into the camera images. Deformable attention mechanisms are then applied to amalgamate features from the images, thereby enriching the radar features. Concurrently, a learnable control variable γ is harnessed to adjust the weighting of the image-derived features. Subsequently, the augmented radar features are connected with the pristine radar features, culminating in the acquisition of integrated radar features.

the sampling offset and attention weight, respectively, for the k-th sampling point within the m-th attention head. They are obtained through the linear projection of Q_i and are used to dynamically adjust the position and weight of the sampling points. These symbols collectively contribute to the feature aggregation process, ensuring that the features extracted from the image effectively enhance the radar features.

In adverse weather conditions, the quality of camera imaging may be compromised by moisture in the air, resulting in blurred vision and a loss of meaningful image information. Leveraging camera imagery to enhance radar features under these circumstances may lead to a decrease in detection accuracy. Therefore, we introduce a learnable control variable γ to adaptively adjust the weight of camera image features. Finally, the enhanced radar features are concatenated with the original radar features to obtain the fused radar features:

$$F_R^* = F_R + \gamma \times DeformAtten(Q_i, p_i, F_I) \quad (3)$$

where F_R represents the original radar features, and F_R^* denotes the radar features after enhancement. The model structure is shown in Figure 3.

3.2. Geometric transfer

After processing through different backbone networks, the radar and camera images are endowed with spatial geometric and semantic features, respectively. To further integrate these features into a unified BEV space for robust object detection, it is necessary to perform a view transformation on the images captured by the camera, projecting the 2D image features into the camera BEV space (Y. Liu et al., 2023; Yang et al., 2023).

To project camera image features into the BEV space, we adopt the Lift-Splat-Shoot approach (Philion & Fidler, 2020), which lifts 2D image features $F_I \in \mathbb{R}^{C\times H\times W}$ into a 3D volume by predicting per-pixel

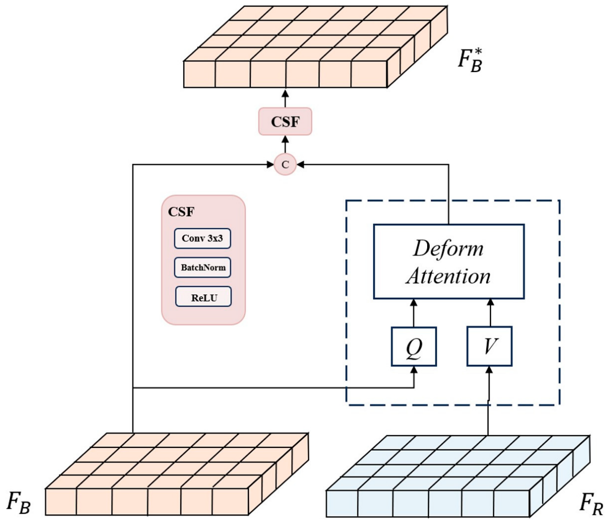

Figure 4. Geometric Transfer module. Based on the initial BEV features generated from the camera images, we optimise the spatial distribution of the camera image BEV features using the BEV features derived from radar, employing a deformable attention mechanism. Subsequently, channel and spatial fusion layers are utilised to adaptively aggregate the features.

depth distributions and applying camera parameters. These 3D features are then collapsed into the BEV plane, resulting in features $F_B \in \mathbb{R}^{C\times h\times w}$, which have the same shape as the radar features F_R. However, due to the lack of reliable depth supervision, this view transformation may yield suboptimal BEV features, especially under adverse weather conditions. To address this limitation, we introduce geometric guidance from radar BEV features, which contain more accurate spatial information, to enhance the projection quality of camera BEV features.

Figure 4 illustrates the architecture of the Geometric Transfer module. Building upon the camera images that have explicitly estimated the varying depth probabilities of pixel features and projected them into the BEV space, the Geometric Transfer module further utilises a deformable attention mechanism. Guided by the geometric features of the radar BEV, it optimises the spatial distribution of the camera image BEV features. Specifically, this module takes the original camera BEV features F_B as query, and the radar features F_R as key and value, then performs deformable cross-attention to update the camera BEV features $\tilde{F}_B$ (Zhu et al., 2020):

$$\tilde{F}_B = DeformCrossAtten(F_R, F_B) \tag{4}$$

After updating the camera BEV features $\tilde{F}_B$ through deformable cross-attention, we propose a channel and spatial fusion layer (CSF) to aggregate the BEV features. Specifically, the camera BEV features $\tilde{F}_B$ are concatenated with the original radar BEV features F_R, and this concatenated feature is passed through the CSF layer. The CSF consists of Conv 3×3, Batch Normalization, and ReLU activation function in sequence. Two CSF blocks are then applied to further fuse the multimodal features, and finally obtain the fused feature F_B^*. The fusion process can be formulated as follows:

$$F_B^* = \mathrm{CSF}(Concat(\tilde{F}_B, F_R)) \tag{5}$$

3.3. Cross-attention fusion

After passing through the Semantic Transfer and Geometric Transfer modules, we have obtained enhanced radar features as well as more accurate camera BEV features, denoted as F_R^* and F_B^*, respectively. However, there is a timestamp misalignment issue during sensor data collection, which causes the camera-generated BEV features to be assigned to adjacent BEV grids, leading to a misalignment between the radar and camera BEV features (Jiao et al., 2023; Ye et al., 2023). To mitigate this issue, we employ a cross-attention mechanism to dynamically align the multimodal features (Alexey, 2020; Zhu et al., 2020). Considering that the radar has more accurate spatial information, we use the enhanced radar features F_R^* as query and the camera BEV features F_B^* as key, value to dynamically fuse the features of the two modalities, resulting in the final fused representation F_{final}:

$$F_{final} = DeformCrossAtten(F_R^*, F_B^*) \tag{6}$$

The integrated features from both the radar and camera modalities are subsequently passed to the subsequent detection head, where the object detection is performed to achieve the ultimate detection outcomes.

4. Experiments

4.1. Experimental setup

4.1.1. Dataset

We evaluated our approach on the challenging Radiate dataset (Sheeny et al., 2021), which comprises a total of 61 sequences with high-resolution radar imagery designed to understand safe autonomous driving scenarios across various weather conditions, including clear skies, nighttime, rain, fog, and snow. The driving scenarios vary from highways to urban settings. The data format consists of radar images generated from point clouds, where pixel values represent the intensity of radar signal reflections. Although the dataset offers high-quality radar data, the quality of its camera and LiDAR data does not match other autonomous driving datasets (Caesar et al., 2020; Geiger et al., 2012), such as the Waymo Open Dataset (Sun et al., 2020). However, this limitation makes the robustness assessment of our proposed sensor fusion algorithm more remarkable. In all our experiments, we trained the model on a training set that includes both favourable and adverse weather conditions and evaluated the resultant models on a standard test set.

4.1.2. Implementation details

We implement our network in PyTorch using the open-sourced Detectron2. For both the image and radar branches, the backbone is ResNet-50, initialised using weights from a ResNet-50 model pre-trained on ImageNet (Deng et al., 2009). The learnable variable γ introduced in the Semantic Transfer module is initialised to a small positive value to ensure that the model initially relies more on radar features, as radar features are generally more stable under adverse weather conditions. During training, γ is updated through backpropagation, and its gradient is computed based on the difference between the predicted results and the ground-truth object detection outcomes. In this way, the contribution of the camera image features can be dynamically adjusted according to their reliability under different weather conditions.

The input camera images are of a resolution 672×376, while the radar images are captured at a higher resolution of 1152×1152. For the Semantic Transfer section, the decoder has 4 layers. In the Geometric Transfer section, to transform the camera view onto the BEV plane, we set the distance range in both the x and y directions from 0 to 50 m, with each grid cell measuring 0.5 m by 0.5 m. For the depth range predicted by the network, we established it to be between 4 and 45 m, with an interval of 1 m, the decoder in this part has two layers. Finally, in the Cross-attention Fusion component, the decoder consists of two layers as well. We used the Adam optimiser (Diederik, 2014) to complete all training on a single GTX 3090, with a batch size of 2, over the course of 35 epochs, using a learning rate of 2.5e−4 and a weight decay of 1e−2.

4.1.3. Evaluation indicators

We reported the BEV mAP at an IoU threshold of 0.5, aligning with the baseline recommendations suggested within the Radiate dataset. The baseline is a Faster R-CNN detector (Ren et al., 2016) with ResNet-50 (He

Table 1. Experimental results of object detection on Radiate dataset.

	Split: train good weather 2	Split: train good and bad weather
Baseline	45.31	45.77
DRCFusion-R	48.92	49.32
DRCFusion	**53.94**	**54.98**

Table 2. Ablation of the main components of DRCFusion.

Method	ST	GT	CAF	mAP
(a)				50.05
(b)	✓			50.88
(c)		✓		51.83
(d)			✓	52.73
(e)	✓	✓	✓	**53.94**

et al., 2016) as the backbone. Here are the calculation formulas:

$$\begin{cases} P = \dfrac{TP}{TP + FP} \\ R = \dfrac{TP}{TP + FN} \\ AP = \displaystyle\int_0^1 P(R)\,\mathrm{d}R \\ mAP = \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} AP_i \end{cases} \tag{7}$$

where TP represents true positives, FP represents false positives, and FN represents false negatives.

4.2. Main results

We evaluated the performance of our proposed method on the Radiate dataset (Sheeny et al., 2021) and summarised the results in Table 1.

Our DRCFusion-R, which is based solely on radar sensors, achieved results that outperformed the baseline by 3.61 percentage points when trained on the train good weather dataset. When trained on the train good and bad weather dataset, the performance improvement over the baseline was around 3.55 percentage points, which is also 0.4 percentage point higher than the results from the train good weather dataset. Intuitively, since the test set encompasses data from a variety of weather conditions, incorporating a more diverse range of weather data during training can enhance the final detection outcomes as well as the robustness of the detection.

Our proposed fusion model, DRCFusion, equipped with the Semantic Transfer module for enriching radar features and the Geometric Transfer module for assisting in the projection of camera features onto the BEV plane, has demonstrated a significant improvement in BEV mAP over the baseline by 8.63 percentage points under train good weather conditions, and by 5.02 percentage points over the radar-only DRCFusion-R. Under train good and bad weather dataset, the model outperformed the baseline by 9.21 percentage points and improved upon DRCFusion-R by 5.66 percentage points. These gains substantiate the effectiveness of our proposed sensor fusion model.

In the forthcoming sections, we will examine the performance of this method in more detail.

4.3. Ablation studies

4.3.1. Effect of different DRCFusion modules

To gain insights into how each module within DRCFusion impacts detection performance, we conducted an ablation study on the Radiate dataset. The experimental results are presented in Table 2. This ablation study is crucial for understanding the contribution of individual components within our proposed DRCFusion framework. By systematically removing or altering each module, we can isolate the effects on the overall detection performance, providing valuable information for further refining and optimising our approach.

In method (a), the Semantic Transfer (ST) and Geometric Transfer (GT) modules were discarded. Instead, a simple concatenation of dimensions was employed within the camera image and radar fusion module, followed by a 3×3 convolution and channel attention for feature fusion. The final experimental mAP achieved was 50.05%.

Method (b) expanded upon (a) by integrating the camera image to enrich the radar features through the Semantic Transfer module, resulting in a 0.83% improvement in mAP.

Method (c) introduced the Geometric Transfer module, which improved the geometric features of the camera in the BEV perspective, leading to a 1.78% increase in mAP.

Method (d) employed a cross-attention fusion (CAF) mechanism in the camera image and radar fusion process, dynamically integrating features from both modalities, which resulted in a 2.68% increase in mAP.

Method (e) represents our DRCFusion approach. By integrating all components, we achieved a 3.89% gain in mAP compared to Method (a).

4.3.2. Effect of semantic transfer module

The robustness of our proposed DRCFusion in adverse weather conditions is augmented by the Semantic Transfer module, which dynamically modulates the contribution of camera image features in response to different weather scenarios. In adverse weather, the imaging quality of cameras is severely impacted; water vapor can obscure the view, resulting in the forfeiture of valuable image data. If camera image information is still used to enrich radar features at this point, it may result in a decrease in the final detection accuracy. Our

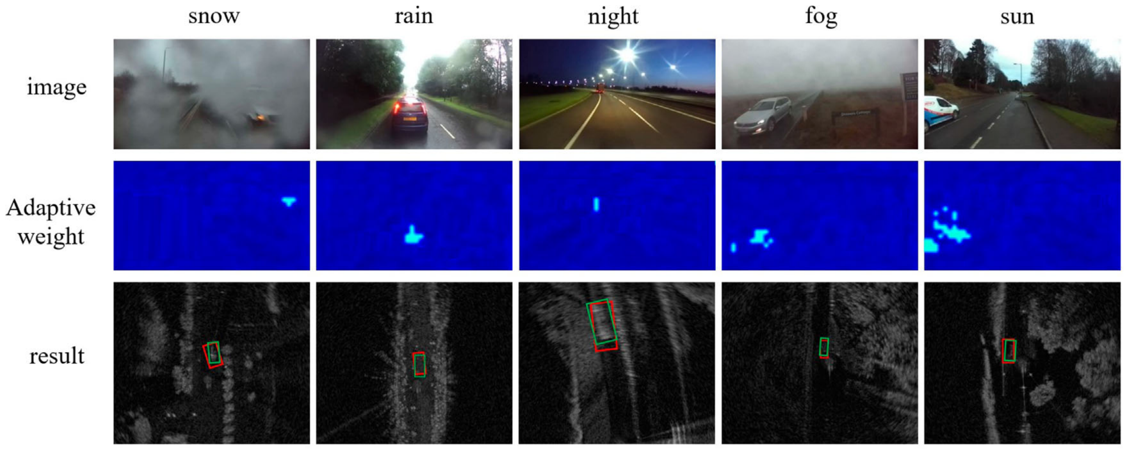

Figure 5. The first row displays the camera-captured images under various weather conditions, the second row illustrates the weight distribution schematic of our Semantic Transfer module's adaptive feature collection from the camera images, and the third row presents the detection results based on our Semantic Transfer module, green bounding boxes are ground-truth annotations, while red are model predictions.

Table 3. Ablation of semantic transfer module.

Method	mAP	snow	rain	night	fog	sun
Baseline	50.05	13.79	28.33	67.95	68.01	55.33
+Base ST	50.77	15.38	32.25	67.87	**70.25**	55.37
+adaptively	**50.88**	**15.85**	**32.83**	**67.97**	70.18	**55.41**

Table 4. Ablation of geometric transfer module.

Method	mAP
Baseline	50.05
+Deformable Cross-Attention	51.50
+Channel and Spatial Fusion	**51.83**

ablation study concerning the Semantic Transfer module is delineated in Table 3. The baseline is maintained as previously established, with the sequential incorporation of a foundational ST module and an adaptively enhanced ST module into our model. The table details the mAP corresponding to the model under adverse weather conditions. Finally, the weight maps collected from camera images by the ST module are shown in Figure 5, illustrating the weight distribution of features collected from the images under different weather conditions.

4.3.3. Effect of geometric transfer module

We conducted an ablation study on the Geometric Transfer module, encompassing both the deformable cross-attention mechanism and the channel and spatial fusion modules, with the results presented in Table 4. The model, which integrates radar BEV features to enhance camera BEV features via cross-attention, experienced a mAP increase from 51.5 to 51.83, thereby corroborating the effectiveness of the cross-modality alignment facilitated by the attention mechanism. Moreover, the incorporation of channel and spatial fusion modules for the amalgamation of BEV features achieved a 0.33 increment in mAP.

5. Conclusion

In this paper, we present DRCFusion, a radar-camera object detector that addresses the challenge of robust object detection under adverse weather conditions. A Semantic Transfer module has been intricately designed to adaptively modulate the significance of camera image features contingent upon the prevailing weather conditions. Additionally, to facilitate the optimal projection of camera images onto the BEV plane during severe weather, the Geometric Transfer module is introduced. This module leverages the spatial features of radar data to implicitly refine the distribution of camera image features within the BEV. Our method's efficacy in object detection has been rigorously evaluated on the radiate dataset, and when juxtaposed with baseline methodologies, our approach has illustrated marked enhancements in performance.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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